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Injectable Weight-Loss Medications

A Policy Analysis

17th February 2025

Commercial Determinants of Health and Health Policy – *Emmanuel, Sara, Yrina, Aljira, & Sian*

1 Introduction

Diabetes, Obesity and Available Medication

Diabetes in the U.K

- Data shows that almost 4.6 million people in the UK live with diabetes
- Nearly 1.3 million people could be living with undiagnosed type 2 diabetes
- An estimated 12.1 million adults in the UK are living with diabetes or pre-diabetes (1)

Obesity in the U.K

62% of the U.K population is overweight or obese (2)





Figures 1 and 2: Obesity rates in the U.K stratified by deprivation and educational level







Available Medication (3)

Semaglutide / liraglutide – GLP-1 agonist & delayed gastric emptying – licensed in 2017 for diabetes and in 2021 for weight loss

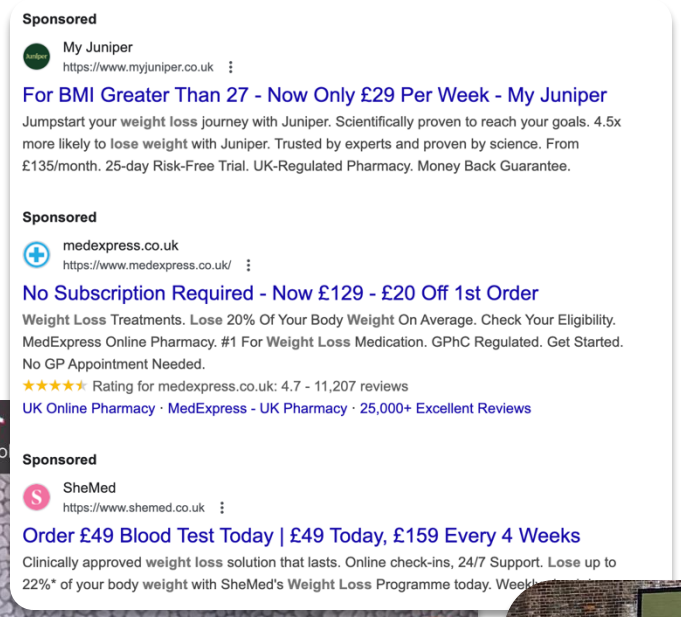
Tirzepatide – increases insulin secretion from the pancreas

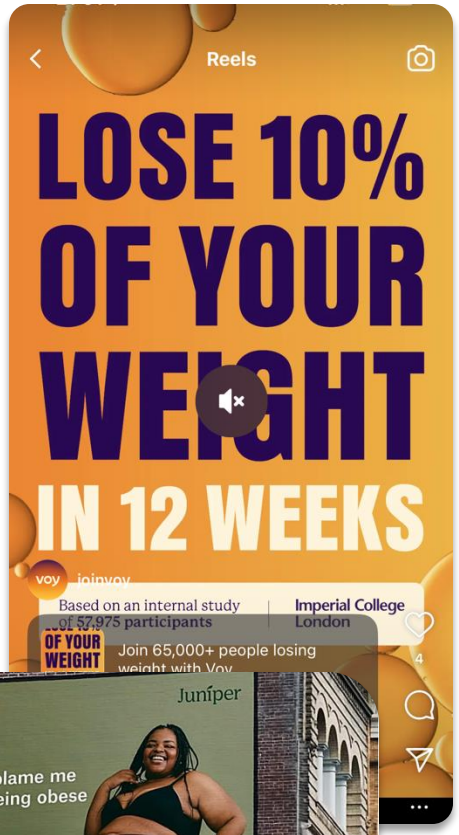
Side Effects

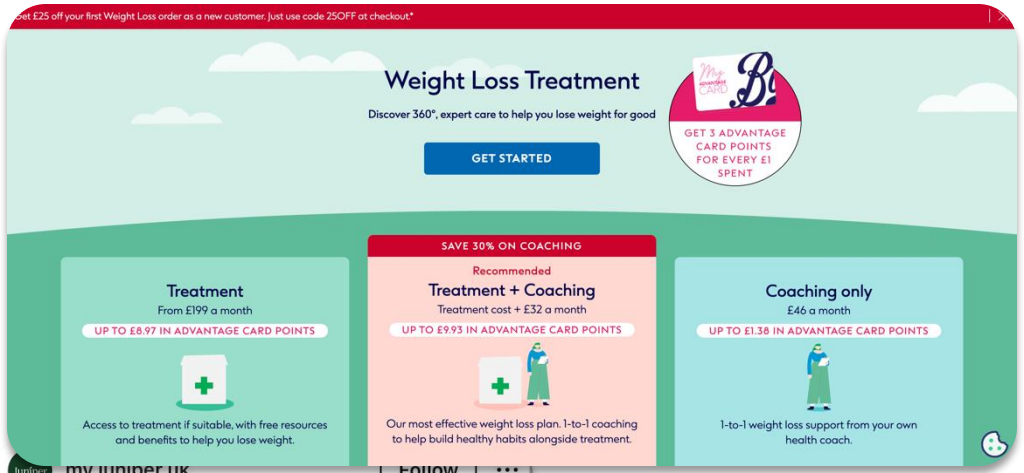
nausea, vomiting, diarrhoea, constipation, bloating, gas, stomach pain, loss of appetite, fatigue, headaches, dizziness, severe vomiting, dehydration, pancreatitis, gallbladder problems, kidney issues, severe hypoglycemia (in diabetics), increased heart rate, thyroid cancer risk (medullary thyroid carcinoma), severe gastroparesis (stomach paralysis), suicidal thoughts, weight regain after stopping, potential cardiovascular risks, fertility and risk to pregnancy

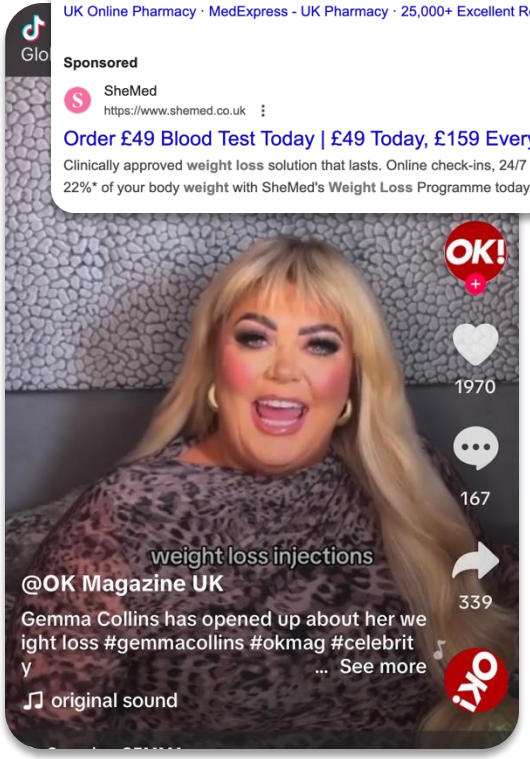
(1) Diabetes UK (2024). How Many People in the UK Have diabetes? *Diabetes UK*. **(2)** Nuffield Trust. (2024). Obesity **(3)** Knudsen, L.B. and Lau, J. (2019). The Discovery and Development of Liraglutide and Semaglutide. *Frontiers in Endocrinology*

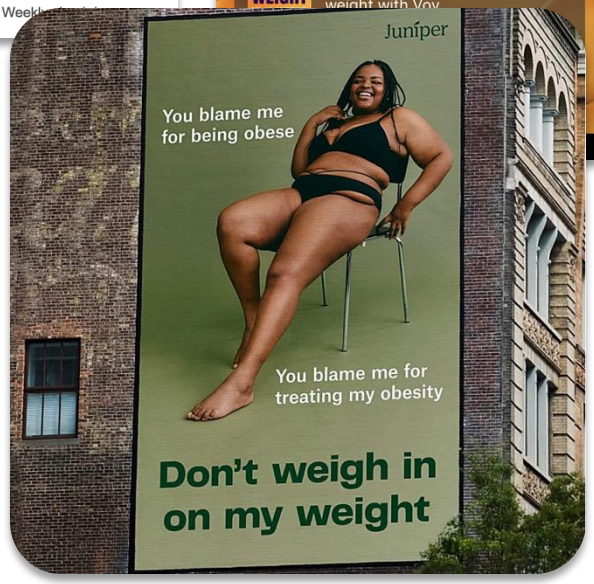
Advertising Across All Platforms

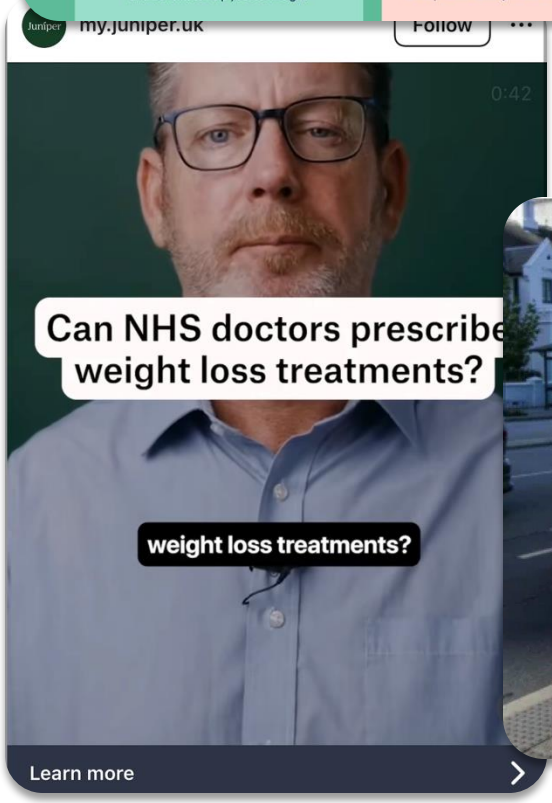












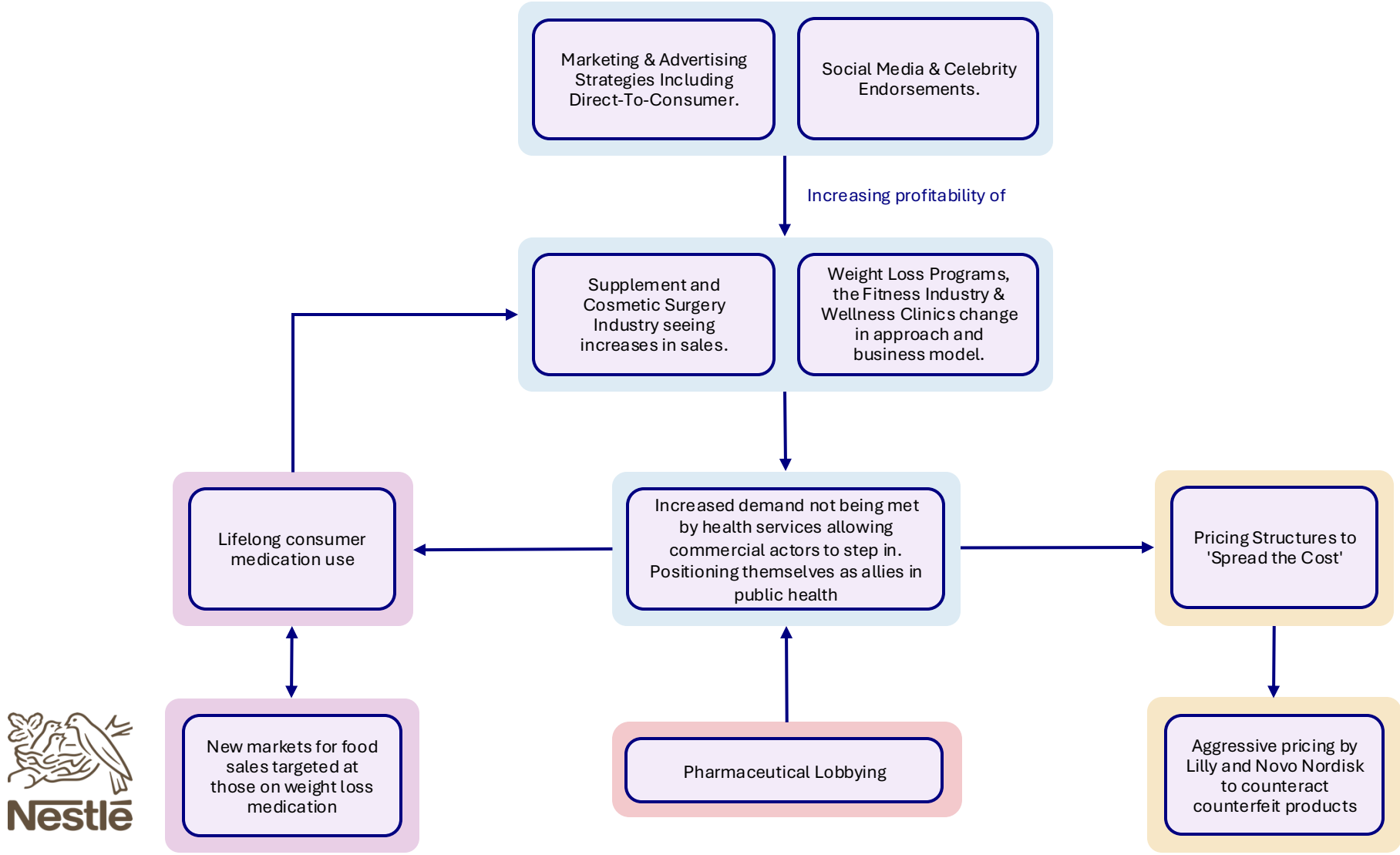


Oh Oh Oh Ozempic



https://www.youtube.com/watch?v=YlF5XrTO6rw&t=5s

A Booming Market and Its Interactions



Remarkable Success of Pharmacological Interventions Could have Wider Public Health Implications

Disease Prevention (4)

- Shift in Focus from Prevention to Treatment: Focus shifts attention away from prevention strategies like promoting balanced diets and regular physical activity.
- Overemphasis on Medication: Undermine long-term public health efforts to address the root causes of obesity; poor dietary habits, lack of exercise, and environmental factors.

Ongoing Infodemic (5)

- Global search interest is rising over time: Upward search trend usually aligns with media and social media influence. Regulators link shortages to social media-driven off-label demand
- Potential for misuse among individuals without obesity or diabetes: Rise of social media influencers promoting the drug for cosmetic purposes.

Health Equity and Access (6)

- Increased Demand for Prescriptions: The growing popularity of injectable weight-loss medications could strain healthcare systems, with more individuals seeking prescriptions not only for diabetes management but also for weight loss.
- Risk of Worsening Health Inequality: Limit access for individuals who need it for diabetes management, worsening health disparities in those with chronic conditions.

In 2023, The Medicines and Healthcare products Regulatory Agency (MHRA) issued a press release warning the public about potentially fake pens, advising caution when purchasing medicines online 3



Picture of an example falsified pen on top, and a genuine Novo Nordisk pen on the bottom, courtesy of Novo Nordisk.

(4) Ryan, N. et al. (2025). The Ethics of Ozempic and Wegovy. *Journal of Medical Ethics* **(5)** Raubenheimer, J.E. et al. (2024). Sweetening the deal: an infodemiological study of worldwide interest in semaglutide using Google Trends extended for health application programming interface. *BMC Global and Public Health* **(6)** GOV.UK. (2023). MHRA warns of unsafe fake weight loss pens.

- Medications for side effects of ozempic (prescription cascade).

Easy Access with Minimal Medical Oversight

- Potential users can select their preferred medication through a personalised pathway.
- Medication options are offered after inputting postcode, gender, and date of birth, without considering weight or clinical markers.
- Some pharmacies allow users to enter their weight without verification.

Pharmaceuticalisation and Commercial Influence

- Availability of **complementary weight-loss supplements** before prescribing medication, highlighting commercial incentives.
- Verification processes vary—Only Boots requires a full-body picture, selfie, and ID, whereas ASDA does not require weight input at all.
- **Medicalisation** – Obesity is increasingly being treated as a medical condition requiring pharmaceutical intervention, rather than focusing on lifestyle changes. (7)
- **Pharmaceuticalisation** – The minimal medical checks and quick, prescription-based solutions indicate a growing reliance on medication rather than holistic approaches such as diet and exercise.

- New market for 'natural' weight loss medication eg. GLP-1 tablets



Figure 2: Taken from Boots Online weight-loss injectable prescribing form

(7) Fitzpatrick, M. (2005). Selling Sickness: How Drug Companies Are Turning Us All Into Patients. *BMJ : British Medical Journal*,

2 Policy, Process, & Interest

Timeline of Weight-Loss Injectables in the UK

1 — March 2024

NICE Endorses Semaglutide (8)

NICE recommended semaglutide for adults with BMI of 35 kg/m^2 or higher and at least one weight-related comorbidity

2 — September 2024

Semaglutide Introduced to The NHS

Semaglutide became available through NHS on September 4, 2023. Also includes BMI between 30 and 34.9 kg/m^2 with specific conditions

3 — June 2024

Govt. Initiates Pilot Programs (9)

Government allocated £40 million over two years to pilot the accessibility of new obesity medications, aiming to expand availability beyond hospital settings

4 — October 2024

NICE Evaluates Tirzepatipde

NICE recommended Tirzepatide for adults with BMI of 35 kg/m^2 or higher and at least one weight-related comorbidity

5 — October 2024

Investment in Weight-Loss Medication Trails

£279 million investment to support clinical trials evaluating Tirzepatide's effectiveness in weight loss and obesity-related complications

6 — December 2024

Final Guidance on Trizepatide

NICE proposed a rollout due to the substantial number of eligible patients and concerns regarding NHS capacity. Prioritised highest clinical need – reaching 200,000 individuals over 3 years.

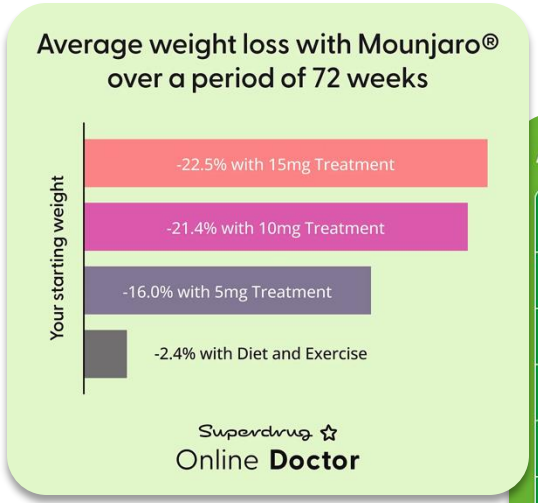
7 — February 2025

Strengthened Regulations for Online Prescriptions (10)

The General Pharmaceutical Council implemented stricter guidelines for online pharmacies. New measures require independent verification

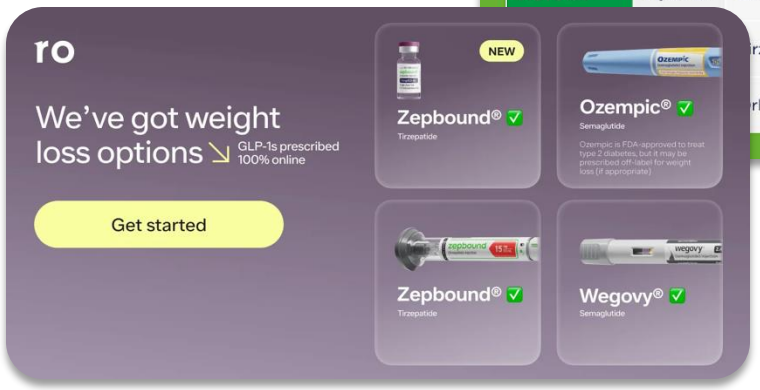
(8) General Pharmaceutical Council. (2024). *Weight loss medications- FAQ | General Pharmaceutical Council*. **(9)** Balogun B. Weight loss medicines in England: House of Commons briefing . *London: House of Commons Library*
(10) Davis, N. (2025). UK online pharmacies face stricter rules for sales of weight-loss jabs. *The Guardian*

Stricter Guidance for Online Pharmacies



ASDA Online Doctor by ZAVA — **Weight loss medication**

Medication	Type	Active Ingredient	How it works	How often to take it
Orlistat	Pill	Orlistat	Prevents fat absorption	3 times daily, with each meal
Xenical	Pill	Orlistat	Prevents fat absorption	3 times daily, with each meal
MySimba	Pill	Bupropion-Naltrexone	Reduces Appetite	Twice daily
Wegovy	Injection	Semaglutide	Reduces Appetite	Weekly
Nevolat	Injection	Liraglutide	Reduces Appetite	Daily
		rzepatide	Reduces Appetite	Weekly
		rlistat	Prevents fat absorption	3 times daily, with each meal



The General Pharmaceutical Council (GPhC)

- Patient safety
- Inappropriate prescriptions
- Potential misuse

New Criteria

- Information must be verified independently
- Person's height, weight, and BMI through video consultation, person's clinical records or by contacting another healthcare provider such as the person's GP

If Legal Requirements are Breached?

- First step: GPhC will give the pharmacies an improvement action plan
- Next steps: Limiting the medications they can supply, suspending a pharmacy, or taking action against pharmacy owners and pharmacists themselves

Actors are individuals or groups that actively influence or shape policy decisions & implementation

Government & Regulatory Bodies

- National Institute for Health & Care Excellence (NICE): Determines prescribing guidelines
- Department of Health & Social Care (DHSC): Shapes national health policy
- Medicines & Healthcare products Regulatory Agency (MHRA): Ensures drug safety
- General Pharmaceutical Council (GPhC): Implements new rules for online prescription
- National Pharmacy Association (NPA): Deals with tougher rules regarding prescription
- General Medical Council (GMC): Regulates medical professionals

Regulatory & Structural Power: Create, enforce, & modify policies that determine drug approval, prescription guidelines, & public health measures

Healthcare Staff & Service Providers

- GPs & NHS Providers: Prescribes & monitors treatments
- Community Pharmacists: Dispenses medication & advises patients
- Private Weight-loss Clinics: Provides non-NHS access to drugs

Expert & Dispositional Power: Directly impact a person's access through prescription, treatment, and dispensing decisions, influenced by guidelines (but with discretion)

Private Sector & Pharma Companies

- Novo Nordisk: Manufacturer of Ozempic (or, Wegovy)
- Eli Lilly: Producer of competing weight-loss injectables
- Online Pharmacies & Retailers: Supplies (un)regulated drugs
- Private Health Clinics: Provides alternate access to drugs

Economic & Structural Power: Control production, pricing, & marketing, influencing both policy and consumer demand through lobbying, funding, framing, & supply chains

Patient & Consumer Groups

- People with Obesity or Diabetes: Direct users of the medication
- Patient Advocacy Organisations: Push for safety & accessibility

Advocacy & Relational Power: Influence public discourse, pressure policy makers for increased access, & spread awareness via lobbying & social movements

Media & Public Influencers

- Social Media Influencers & Celebrities: Drive demand & mis-info
- Health Journalists & Investigators: Highlight policy gaps & risks

Normative & Relational Power: Shape public opinion, consumer demand, & apply political pressure by elucidating policy gaps, promoting trends, or misinformation

Academic & Research Institutes

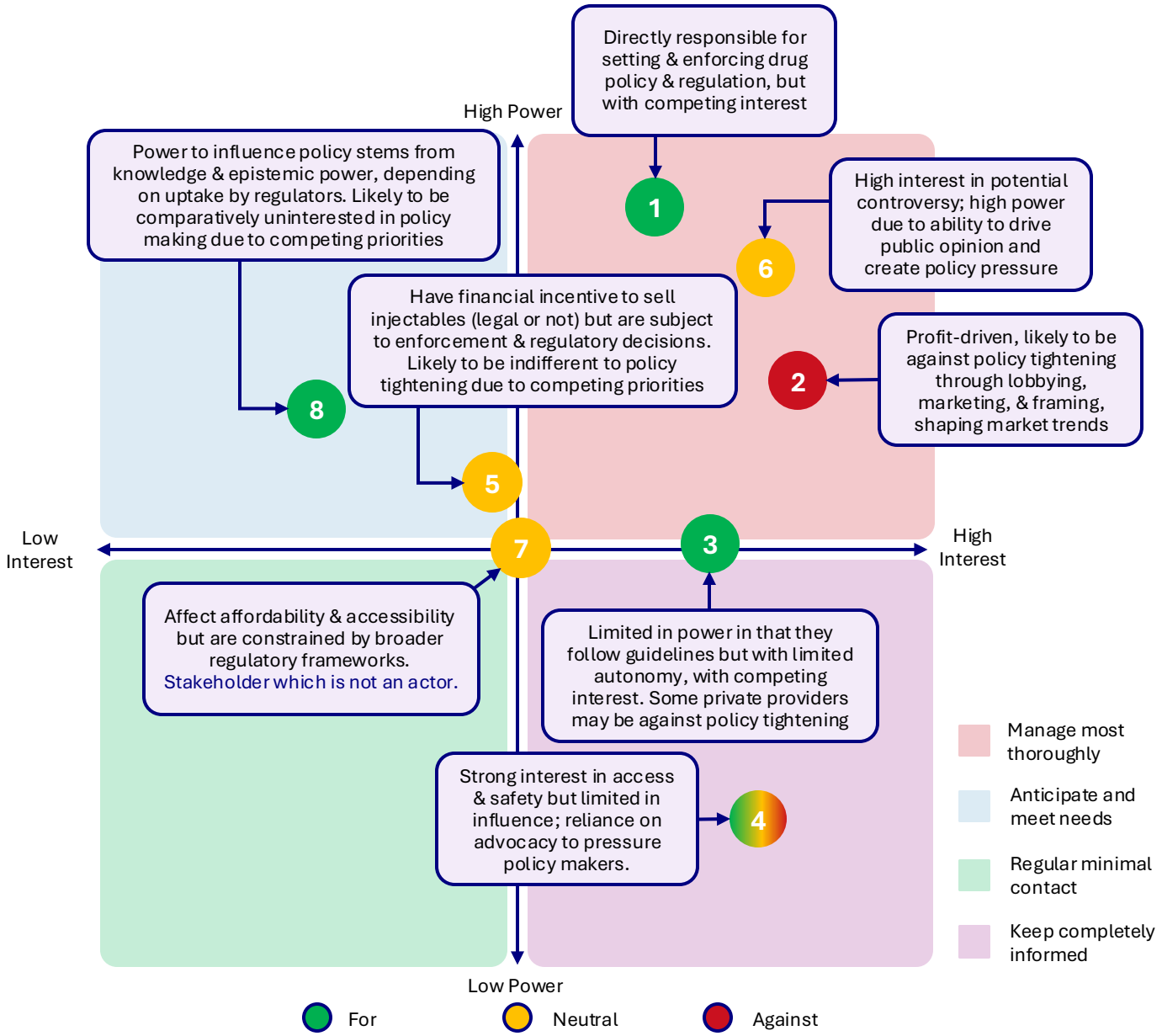
- Public Health & Medical Researchers: Study efficacy, risks, & impacts
- Think Tanks: (i.e. King's Fund), these provide policy recommendations

Knowledge & Epistemic Power: Inform policy decisions through evidence-based research, pharmaceutical innovation, & public health interventions

Understanding Actors

- Actors are individuals, groups, or organizations that actively influence or shape injectable weight-loss drug policy decisions and implementation
- They are directly involved in policy making, regulation, lobbying, or enforcement
- They often have power and agency in shaping these policies and decisions

Stakeholders are individuals or groups with a vested interest in injectables policy outcomes



Understanding Stakeholders

- Stakeholders are slightly different from actors in that they are individuals, groups, or organisations with a vested interest in the policy outcome, whether actively involved or passively affected
- They may benefit from, or be harmed by policy decisions
- They do not always have direct influence over these policy decisions
- They may have varying levels of influence; some are passive recipients of policy outcomes, while others are not
- They have varying degrees of different types of power and interest, and their positions are not stagnant

1. Government & Regulators (NICE, MHRA, etc.)
2. Weight-Loss Injectable Producing Pharmaceutical Companies (Novo Nordisk, Eli Lilly, etc.)
3. NHS & Private Healthcare Providers
4. Patients & Consumer Groups (Diabetes UK, etc.)
5. Online Pharmacies & Retailers
6. Media & Social Media Influencers
7. Insurance Providers
8. Academic & Research Institutions

There are more stakeholders than listed; the above are just a few

Commercial Determinants of Health Framework

Commercial Determinants of Health

The systems, structures and strategies used by commercial actors that shape health outcomes.

According to Kickbusch et al. (13), there are 4 key mechanisms:

1. Marketing and promotion
2. Lobbying and corporate influence
3. Supply chains and trade
4. Corporate social responsibility

Marketing & Promotion

- Aggressive marketing of Ozempic for weight-loss (11).
- Social media influencers and celebrity endorsement, increasing demand outside of medical necessity.
- 'Off-label' use encouraged by some providers.

Lobbying & Corporate Influence

- Novo Nordisk lobbied for expanded access but was suspended from ABPI (2023) for breaching regulations (15).
- Pressure on NICE to expand prescription criteria (14).
- Political and regulatory debate regarding pharma companies influence on health policy (9).

Supply Chains & Trade

- Global shortage due to high demand, affecting diabetic patients.
- Unregulated online sales, including counterfeit versions.
- Semaglutide is included in list of drugs that cannot be parallel exported or hoarded to support supply issues and prioritise NHS patients (9).

Corporate Social Responsibility

- Novo Nordisk positions itself as tackling diabetes and obesity, but profits from high demand.
- Ethical concerns over affordability and equal access.
- Limited transparency on lobbying and marketing strategies.

(9) Balogun B. *Weight loss medicines in England: House of Commons briefing* **(11)** London: House of Commons Library. Davis, N. (2024). *'Wild west': experts concerned by illegal promotion of weight-loss jabs in UK. The Guardian* **(12)** Das, S. et al.. (2023). Revealed: experts who praised new 'skinny jab' received payments from drug maker. *The Observer*. **(13)** Kickbusch I, Allen L, Franz C. The commercial determinants of health. *Lancet Glob Health*. **(14)** Jack A. Drug companies' lobbying of ministers has little effect on NICE. *Financial Times*. **(15)** Grover N. Novo Nordisk suspended from UK lobby group. *Reuters*. 2023

Kingdon's Multiple Streams Framework (MSF)

Kingdon's Multiple Streams Framework

Defines a policy window, i.e. why and when policies change, as the intersection between 2 or more of the following streams.

- Problem stream – identifying issues that require government attention
- Policy stream – proposes potential solutions and evaluates feasibility
- Politics stream – reflects political mood, media attention, stakeholder interests

Alignment of public health risks, policy proposals and political momentum makes the introduction of changes to Ozempic regulation more probable.

Problem

Health risks: Increased number of people buying Ozempic online due to strict NICE eligibility criteria.

Consequences: reports of fake/counterfeit medication, misuse, lack of medical supervision.

Equity concerns: restrictive eligibility criteria may exclude some who could benefit.

Healthcare costs: potential consequences of long-term or unmonitored use, creates additional costs and NHS burden.

Policy

NICE current guidelines create eligibility criteria, for patients with BMI≥35 or lower with comorbidities.

Regulatory weakness: MHRA struggles to control and regulate online sale of weight-loss drugs, increased use from unverified sources.

Potential solutions: expand prescription criteria, stronger pharmaceutical regulations, improved supervision., public education on safe use (9).

Politics

Growing public and media attention on weight-loss injectables

Increased advertising of weight-loss injectables, including from non-reputable suppliers.

Novo Nordisk suspended from UK lobby group Association of British Pharmaceutical Industry (ABPI) in 2023 for breaches of code of practice (15)

Government and NHS pressure: balancing access and demand for weight-loss drugs vs concerns over cost and safety.

(9) Balogun B. *Weight loss medicines in England: House of Commons briefing* **(15)** Grover N. Novo Nordisk suspended from UK lobby group. *Reuters*. 2023

3 Approaches from Other Countries & Recommendations

1 Australia (16)

- Semaglutide shortage in early 2022.
- Many Australians with type 2 diabetes struggled to access Ozempic and urged government action on its social media exposure.
- Prescription-only medicine cannot be advertised to the public in Australia.
- Media broadcasters were reminded of their obligations when reporting on therapeutic goods.



2 Norway (17)

- Ozempic consumption sharp rise in 2023.
- Off-label demand caused supply shortages, leading to reliance on costly foreign supply.
- The Norwegian Medical Products Agency (DMP) strongly warned against off-label Ozempic use and reminded doctors not to prescribe Ozempic under reimbursement for obesity treatment.



3 Denmark (18)

- Home of Novo Nordisk, Ozempic production contributed significantly to Denmark's economic growth.
- Sharp increase in use 2021–2022.
- According to a nationwide population-based study, 12% of users in 2023 did not have type 2 diabetes, despite no approval for weight loss and 33% of new users in 2022 did not have type 2 diabetes at initiation



- Increased demand and supply challenges.
- Off-label use concerns.
- Regulatory and economic implications.

(16) Australian Government. Ozempic (semaglutide) shortage 2022–2025. Canberra: *Therapeutic Goods Administration* **(17)** Durvy B. The off-label use of Ozempic led to increased health spending and concerning shortages of diabetes medicine. *Health Systems and Policy Monitor* **(18)** Mailhac A, et al. Semaglutide use in Denmark 2018 through 2023 – user trends and off-label prescribing for weight loss. *Clin Epidemiol.*

Recommendations for Policies and Regulation

Current medicine advertising rules:

- Imply that seeing a doctor or pharmacist is not necessary.
- quote recommendations by scientists, healthcare professionals or celebrities.

Advertising	Enforce existing regulation – appropriately proportionate fines.
Public Health	Promote behavioural change focusing on sustainable lifestyle modifications and raise public awareness about danger of unregulated purchases.
NICE	Regulation on funding & evidence for producing guidelines.
Monitoring	Active monitoring & reporting of adverse events & reactions.

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Thank You

Questions are Welcomed

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